

Laboratory sessions

Laboratory assignment 6

Introduction to Catapult



Goals

- ❑ Introduction to the description of systems using the algorithmic C libraries
- ❑ Introduction to the Catapult high-level synthesis tool
- ❑ Analyze the synthesis results using the Design Analyzer tool
- ❑ Run automatic verification of the C++ and resulting RTL descriptions

Code (1/2)

❑ mult_add_pipeline_ref.cpp

```
#include "mult_add_pipeline_ref.h"
void mult_add_pipeline_ref(unsigned int    a,
                           unsigned int    b,
                           unsigned int    c,
                           ac_fixed<10,2,false> gain,
                           bool            gain_adjust,
                           unsigned int    &result)
{
    int product = a * b;
    int sum = product + c;
    if (gain_adjust) {
        sum = sum*gain.to_double();
    }
    result = sum<<4; // multiply by 16
}
```

Code (2/2)

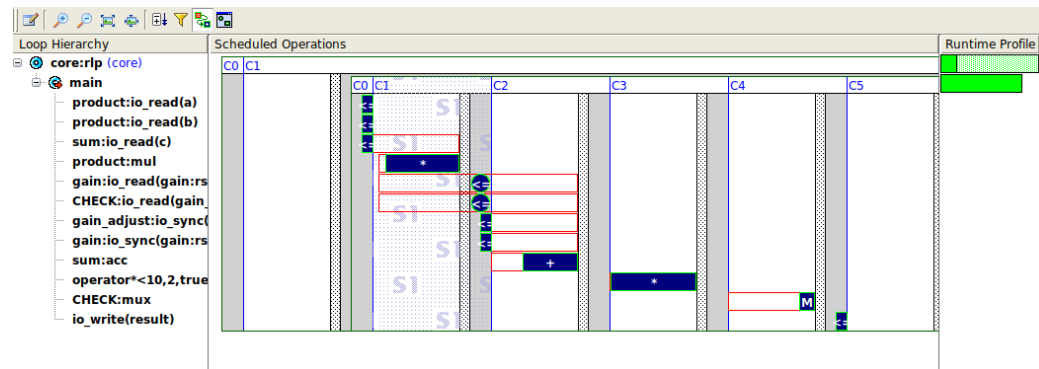
❑ mult_add_pipeline.cpp

```
#include "mult_add_pipeline.h"
#include <mc_scverify.h>
#pragma hls_design top
void CCS_BLOCK(mult_add_pipeline)(ac_channel<ac_int<11,false> > &a,
                                ac_channel<ac_int<14,false> > &b,
                                ac_channel<ac_int<25,false> > &c,
                                ac_fixed<10,2,true>          gain,
                                bool                          gain_adjust,
                                ac_channel<ac_int<30,false> > &result)
{
    ac_int<25,false> product = a.read() * b.read();
    ac_int<26,false> sum = product + c.read();
    CHECK:if (gain_adjust) {
        sum = ((ac_fixed<26,26,false>)(sum*gain)).to_uint();
    }
    // NOTE: cast sum to ac_int<30,false> to avoid losing precision on left shift to multiply by 16
    result.write((ac_int<30,false> )sum<< 4);
}
```

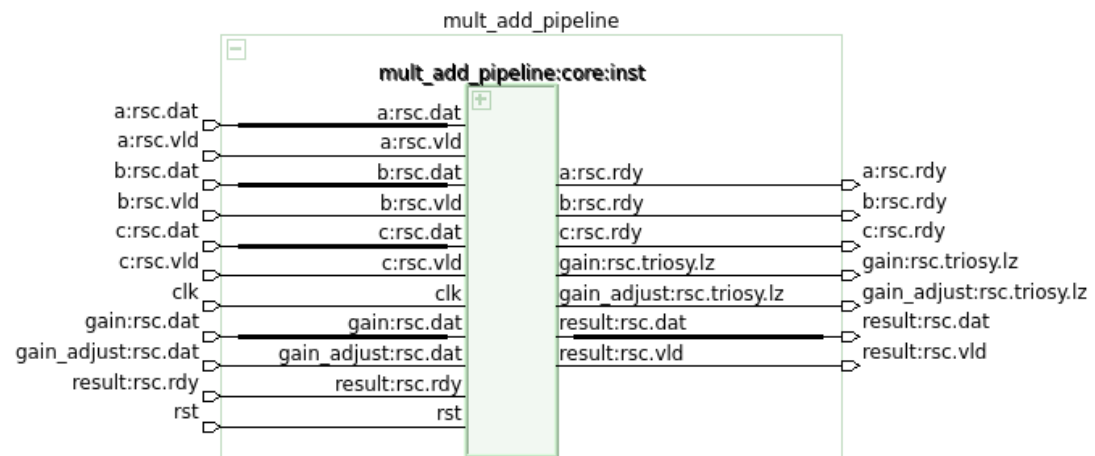
Simple example

- Reference design (C++): mult_add_pipeline_ref.cpp
- Synthesizable design (AC): mult_add_pipeline.cpp

Scheduling

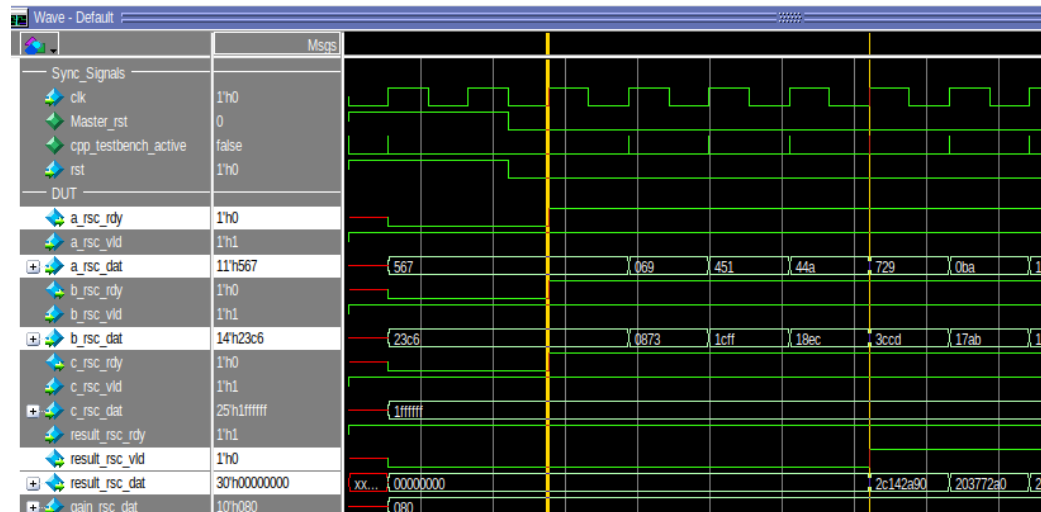


I/O interface



Simple example

- ❑ Automatic verification of the C++ vs. RTL
 - QuestaSim simulation: Check the latency.



- Redo it for Intel/Altera CycloneV technology.
- What is the maximum frequency you can achieve?